M'HAMED BOUGARA UNIVERSITY OF BOUMERDÈS

FACULTY OF SCIENCES, DEPARTMENT OF COMPUTER SCIENCE

Mini-Project Report

UMBB MobGuide

Digital Guide of the University of Boumerdès

Specialty: L3-SI
Academic Year: 2025–2026 (S6)
Module: AM

Project Developed By:

Name	Group	Student ID
AYACHE Melissa	Group 3	232331208118
AZOUAOU Toufik	Group 5	232331533008
BENDOU Maysa	Group 2	232331210302
NAIT SAIDI Lydia	Group 2	232331109315
RABEHI Mohamed Sami	Group 8	232331453109
SLIMANI Abd El Djalil	Group 8	232331413910

1 Introduction

UMBB MobGuide is a mobile application developed using Android Studio (Java/XML), designed as a comprehensive digital guide for the University of Boumerdès (UMBB). The application enables users to browse faculties and departments, view detailed information, and easily access contact details and locations for each entity, including descriptions, specialties, and contact information.

2 Application Features

The UMBB MobGuide mobile application incorporates all the required features outlined in the project description, along with optional functionalities and additional improvements that enhance the user experience.

2.1 Implemented Features

- **UMBB Presentation:** A general description of the University of Boumerdès (UMBB) is displayed on the Home page.
- **Faculties List:** A list of all UMBB faculties with their contact details and locations.
- **Departments List:** Lists departments associated with each faculty.
- **Department Details:** For each department, users can view the name, description, specialties, head, contact information, and location.
- **Contact Options:** Faculties and departments include buttons for direct contact (Phone call, SMS, Email) and location via Google Maps.
- **Global Contact Page:** Displays central university contact information (phone, email, website, location).

2.2 Implemented Optional Features

- **Search Functionality:**
  - Search faculties within the faculties list.
  - Search departments within the department list.
- **Multilingual Interface:** Users can toggle between English and Arabic, with language preferences saved for future use.

2.3 Non-Implemented Features

All required and optional features have been implemented, with no features left out.

2.4 Additional Features

- **Splash Screen:** A smooth transition to the Home page with a startup screen.
- **Faculties Shortcut on Home Page:** Quick access buttons to navigate directly to the faculties section.

- **Statistics Section:** Displays key information like the number of faculties, departments, and UMBB's founding year.
- **Bottom Navigation Bar:** Provides quick access to the Home, Faculties, and Contact sections across all main screens.

3 Application Components

3.1 Screens (Activities)

Class	Associated Layout	Description
`WelcomeActivity.java`	`welcome.xml`	The initial screen shown to the user, displaying the app logo and a "Get Started" button that redirects to the Home.
`HomeActivity.java`	`activity_home.xml`	Main home page with UMBB presentation, faculty shortcuts, language toggle, and a BottomNavigationView. Tapping a faculty opens its associated department list.
`FacultyListActivity.java`	`activity_faculty_list.xml`	Displays all faculties with search bar and navigation to departments.
`DepartmentListActivity.java`	`activity_department_list.xml`	Displays departments of a selected faculty with a summary card at the top and contact buttons at the bottom.
`DepartmentDetailActivity.java`	`activity_department_detail.xml`	Provides detailed information for a department, including name, description, head, specialties, and contact actions (Call, SMS, Email, Maps).
`ContactActivity.java`	`activity_contact.xml`	The university's contact page, hosting the ContactFragment inside a FrameLayout.

3.2 Fragments

Class	Associated Layout	Description
HomeFragment.java	fragment_home.xml	Displays the university's general description on the home screen.
FacultyListFragment.java	fragment_faculty_list.xml	A reusable faculty list with a search feature, updating automatically when the language changes through LiveData.
ContactFragment.java	fragment_contact.xml	Displays UMBB contact details and provides external action buttons (Maps, Call, SMS, Email, Website).
BaseFragment.java	No layout	A parent fragment providing shared logic for the language toggle to avoid code duplication.

3.3 Data Models

These classes represent the core data objects of the application.

Class	What it Stores
Faculty.java	Stores information about a faculty, including code, name, dean, email, phone, address, description, map link, and its list of departments.
Department.java	Stores information about a department, including name, head, email, phone, description, and a list of specialties.

3.4 Data Layer

These classes handle data reading and management for the application.

Class	What it Does
UniversityXmlParser.java	Parses the `assets/university.xml` file using `XmlPullParser`, loading data based on the selected language (English or Arabic).
DataProvider.java	Provides cached data (`getFaculties()`) and avoids parsing XML repeatedly.

3.5 Adapters

Adapters link the data lists (`ArrayList`) to `ListView` components.

Class	What it Does
FacultyAdapter.java	Fills the faculty `ListView`, displaying faculty cards with images, names, deans, contact details, and the department count.
DepartmentAdapter.java	Fills the department `ListView`, displaying department names, descriptions, and specialty counts.

3.6 Utility Classes

These classes provide essential application functionalities.

Class	What it Does
LanguageManager.java	Manages the user's language preference, saving it in SharedPreferences and applying the correct locale to each screen so text and data reload in the selected language.
AppViewModel.java	Manages LiveData for the current language and the list of faculties.
CrashReporterApp.java	Catches unexpected crashes and saves error details to a log file on the device for debugging purposes.

3.7 Layout Files (XML)

Each screen and list item is associated with an XML layout file located in the `res/layout` folder.

File	Purpose
welcome.xml	Layout of the welcome screen.
activity_home.xml	Layout of the home screen with faculty list.
activity_faculty_list.xml	Layout of the faculty list with a search bar.
activity_department_list.xml	Layout of the department list with a search bar.
activity_department_detail.xml	Layout of the department detail page.
activity_contact.xml	Layout for the contact screen.
fragment_home.xml	Displays the university description (home fragment).
fragment_faculty_list.xml	Faculty list fragment with search functionality.
fragment_contact.xml	Displays global contact information and action buttons.
header_faculty_summary.xml	Header for the department list, displaying faculty photo, name, and description.
footer_faculty_contact.xml	Footer for the department list, displaying faculty contact buttons.
item_faculty.xml	Layout of a single faculty card in the list.
item_department.xml	Layout of a single department row in the list.
item_specialty.xml	Layout of a single specialty row inside the department detail.

3.8 Other Resources

- **university.xml:** The main data file located in `assets/`. It contains all faculty and department data in both English and Arabic.

- **AndroidManifest.xml:** Declares all six activities and the required permissions: `CALL_PHONE`, `SEND_SMS`, and `INTERNET`.
- **res/values/arrays.xml:** Defines unique accent colors for each faculty, used as the color stripe in list items.
- **res/values/strings.xml & res/values-ar/strings.xml:** Contain all UI text strings in English and Arabic.
- **res/drawable/:** Contains faculty images, icons, and UI shapes such as badges and selectors.

4 Work Distribution

- **AYACHE Melissa:** Developed the Faculty List screen (`FacultyListActivity.java`, `activity_faculty_list.xml`), the Department List screen (`DepartmentListActivity.java`, `activity_department_list.xml`), and the Department Detail screen (`DepartmentDetailActivity.java`, `activity_department_detail.xml`), including the custom list adapters for both views.
- **AZOUAOU Toufik:** Collected all university data from the UMBB official website (faculty descriptions, department information, contacts, and GPS locations) and structured it into `university.xml`. Managed the GitHub repository (branches, merges, version control). Produced and edited the demonstration video. Contributed to UI/UX improvements across multiple screens.
- **BENDOU Maysa:** Led the initial architecture planning of the application (screen flow, component structure, and data organization). Assisted with the Figma design and prepared the project report.
- **NAIT SAIDI Lydia:** Designed all application screens in Figma, defining layout, color scheme, typography, and component structure. These mockups served as the visual reference for the development team during implementation.
- **RABEHI Mohamed Sami:** Implemented the real-time search feature for both faculties and departments. Developed the university Contact page (`ContactActivity.java`, `fragment_contact.xml`), and managed the bottom navigation bar (`bottom_nav_menu.xml`), handling screen routing between Home, Faculties, and Contact. Built the English/Arabic language switch using `LanguageManager.java` and `AppViewModel.java`, with locale persistence via `SharedPreferences`.
- **SLIMANI Abd El Djalil:** Developed the Splash/Welcome screen (`WelcomeActivity.java`, `welcome.xml`) and the Home screen (`HomeActivity.java`, `activity_home.xml`), including the faculty shortcut cards and bottom navigation bar setup.

Shared tasks (AYACHE, AZOUAOU, RABEHI, SLIMANI): All four members jointly handled application integration, connecting screens via `Intents`, linking data across components, resolving merge conflicts on GitHub, coordinating task dependencies, and performing final testing and bug fixes across the full application.

5 Conclusion

The UMBB MobGuide application was successfully developed with all required and optional features. This project allowed us to apply key Android development concepts such as activities, navigation, data handling, and multilingual support, while also integrating native features like calls, SMS, email, and maps. The clear separation between UI, data, and logic ensured a well-structured and maintainable application.

6 Project Resources

GitHub Repository: https://github.com/toufik-az/UMBB-MobGuide.git

Figma Design: https://www.figma.com/design/RG604hd92wr04cTj5KWOc8/AM-MobGuid?node-id=0-1&t=o9AcE2xaTg2Prffq-1